Lecture 9(a): Flexibility needs assessment

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Agenda

- Flexibility classification
- What is fairness?
- Notions of fairness

Flexibility needs assessment

- Optimization problem from the system operator perspective
- Objective is to quantify flexibility needed to avoid probable network incidents
- EU-mandated assessment under Electricity Market Design reform (Reg. 2019/943, amended 2024)
- Defines data & methodology for TSOs/DSOs to quantify national flexibility needs (5–10 yr horizon)

Source: (ENTSO-E & EU DSO Entity), Flexibility Needs Assessment Methodology'.

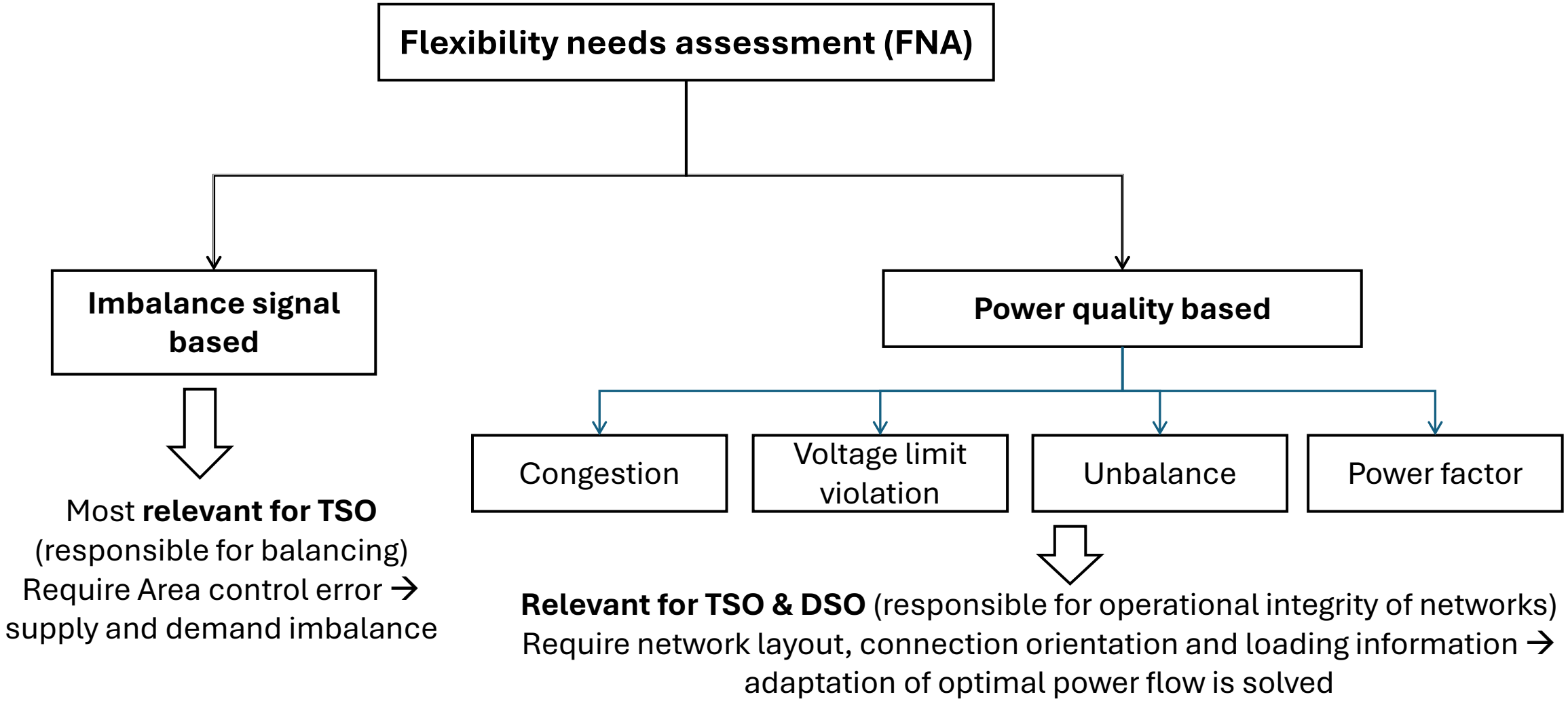
Why flexibility need is increasing

- More variable RES → surplus & curtailment risk
- Steeper residual-load ramps (hourly/diurnal)
- Forecast errors (DA→RT) and forced outages
- Distribution-level congestion & voltage issues

Scope of needs covered with FNA

- System needs: RES integration, ramping, short-term forecast error
- Network needs: DSO congestion/voltage; TSO congestion & balancing, ancillary services
- Technologically neutral metrics: capacity (MW), energy (MWh/TWh), duration (h)

FNA for TSO and DSO



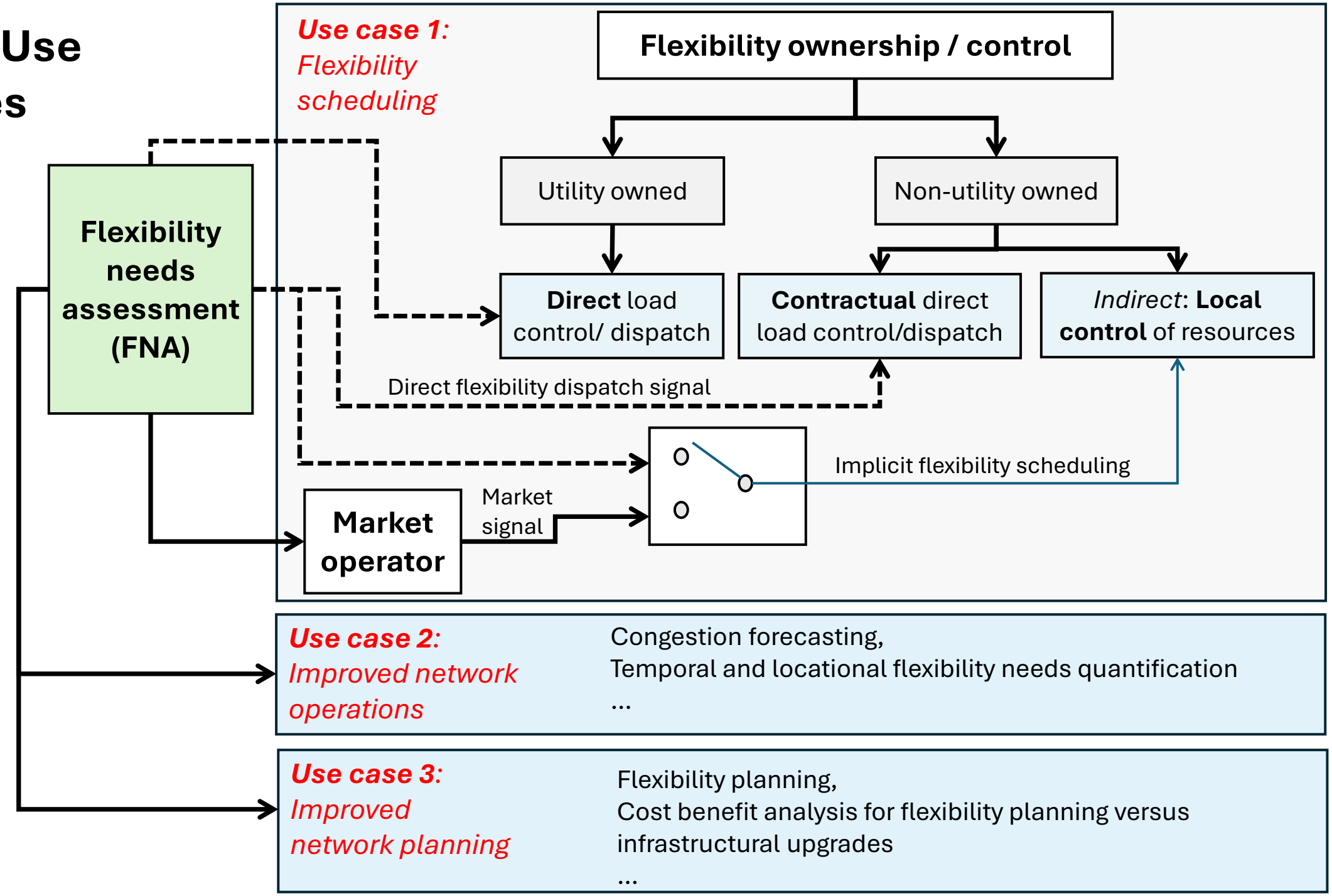
Data requirements and horizons

- TSO inputs aligned with European Resource Adequacy Assessment (ERAA)/National Resource Adequacy Assessment (NRAA) (hourly time series, technical constraints)
- DSO inputs from Distribution Network Development Plan (DNDP); aggregation allowed where detailed data is not meaningful
- Scenarios aligned with national policy targets; climate years as per ERAA/NRAA

Defining system needs

- System need #1: RES integration
- System need #2: Ramping (hourly residual-load swings)
  - Determine hourly residual-load change; compare with real ramp limits, Pmin/Pmax, startup/shutdown.
  - Account for cross-border residual capacity; compute "capacity gap" (MW) up/down.
  - Map frequency, duration, time-of-day (heatmaps/percentiles).
- System need #3: Short-term flexibility (forecast error)
  - Build forecast-error distributions for wind, solar, load (DA→RT).
  - Compute needs at 5 min–5 h; compare to residual ramping margins.
  - Identify hourly capacity gaps and visualize (histograms/violin plots).
- System needs #4: DSO flexibility network needs
  - Each DSO provides 5–10 yr outlook: capacity (MW), energy (TWh), duration, location, voltage level.

FNA Use cases



Elements of the flexibility needs assessment

Required timeframes

The timeframes of flexibility needs span a broad time horizon from "at least" seasonal to daily and hourly flexibility, in particular with regard to RES integration

Scenario-based approach

Scenario-based approach likely needed to account to the different projections for market prices and evolution of generation and demand

Existing & planned flex

Shall consider **"all existing and planned investments,** including existing assets that are not yet connected to the grid" (Rec. 46)

Transmission & distribution

The resources must be considered both on the transmission **and** distribution network levels

Mitigation measures

Part of the report should also focus on evaluating and mitigating barriers to market participation (incl. for system services) of non-fossil flexibility with concrete measures

Cross-border flex

Accounting for potentially available flexibility across borders

Source: What lies behind the (proposed) flexibility needs assessment methodology? Magnus energy

Why FNA for TSO?

Article 9:
System Needs – RES integration

Article 9 identifies how much extra, **technology-neutral, non-fossil flexibility** is required to integrate variable renewables, **in 3 different timeframes**, hourly, daily, seasonal.

It analyses hourly residual-load and curtailment time series across climate and target years to quantify volumes, durations and frequencies of "excess" energy dumped.

By comparing these metrics to a national curtailment target, it then estimates the **MW-and-hours of flexibility needed** to keep RES curtailment within acceptable limits.

Indicator: flexible capacity needed to achieve national RES integration targets.

Article 10:
System Needs – Ramping needs

Article 10 measures the **flexibility required** to follow net-load swings from one hour to the next.

It uses hourly residual-load schedules for generation, storage and interconnectors, factoring in ramp-rate limits, start-up/shut-down constraints and minimum up/down times, to calculate any uncovered upward or downward ramping shortfalls.

The result is an hourly time series and summary statistics that assess when TSOs must secure resources capable of **rapid MW-level adjustments.**

Indicator: additional system needs to cover expected hourly residual load variations

Article 11:
System Needs – Short-term flexibility needs

Article 11 quantifies the reserves needed to cover day-ahead vs. **real-time forecast errors.**

It builds probabilistic distributions of residual-load deviations based on hourly data, defines key percentiles for "slow," "fast" and "very fast" reserves, and overlays these on available asset margins.

The outputs, time series of uncovered shortages plus statistical summaries, guide TSOs in **sizing reserve products by volume and response speed.**

Indicator: additional system needs to cover unexpected variations

Source: What lies behind the (proposed) flexibility needs assessment methodology? Magnus energy

Implications for stakeholders

- TSOs/DSOs: integrate FNA into TYNDP/DNDP; potential roles in scheme implementation.
- Flex providers: new availability-based revenues for DR & storage; technical performance requirements.
- National variations expected; cross-border participation encouraged where CBA positive.

Key Takeaways

- Standardized, technology-neutral FNA identifies system & network flexibility gaps.
- Three core system indicators + DSO network needs underpin targets & support schemes.
- Outputs guide cost-efficient deployment of DR, storage, and grid measures.

Thank you for your attention!

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Heverlee, 08.02.2023

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